PROIECT SQL

Analiza vânzărilor folosind MySQL

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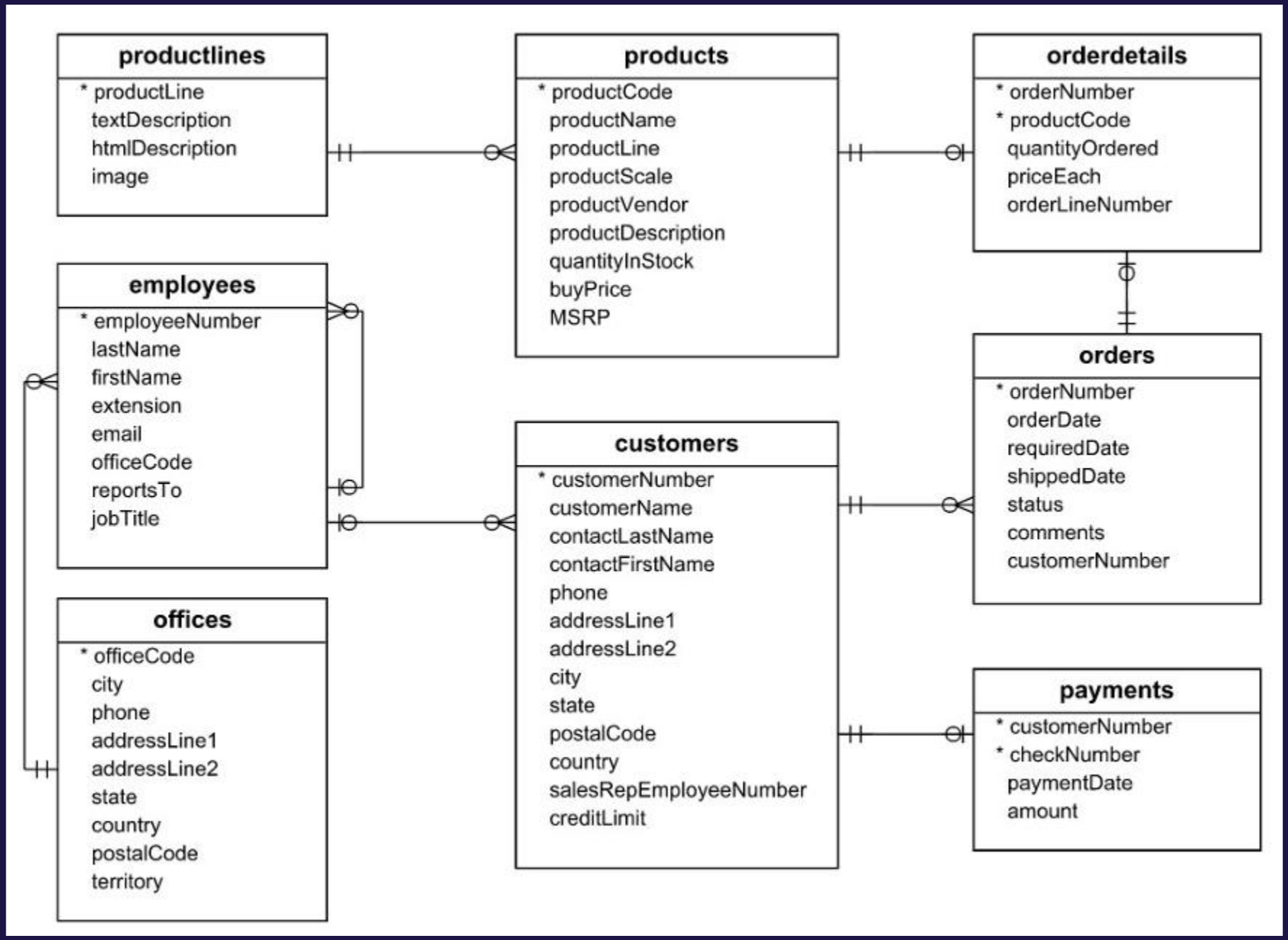
MySQL | 8.0



Prezentarea bazei de date

Classic Models este o bază de date demonstrativă care simulează activitatea unei companii ce comercializează modele de automobile la scară redusă.

Diagrama relațiilor dintre tabele



Scopul proiectului

Utilizarea interogărilor SQL pentru extragerea, agregarea, analiza datelor, identificarea tendințelor și obținerea unor informații relevante pentru procesul decizional.

Obiectiv

Care sunt cele mai vândute 10 produse în funcție de veniturile generate?

Interogare

```sql
SELECT
    od.productCode,
    p.productName,
    SUM(od.quantityOrdered * od.priceEach) AS VenitDinVanzari
FROM orderdetails od
JOIN products p
    ON od.productCode = p.productCode
GROUP BY 1,2
ORDER BY 3 DESC
LIMIT 10;
```

Rezultat

productCode	productName	VenitDinVanzari
S18_3232	1992 Ferrari 360 Spider red	276839.98
S12_1108	2001 Ferrari Enzo	190755.86
S10_1949	1952 Alpine Renault 1300	190017.96
S10_4698	2003 Harley-Davidson Eagle Drag Bike	170686.00
S12_1099	1968 Ford Mustang	161531.48
S12_3891	1969 Ford Falcon	152543.02
S18_1662	1980s Black Hawk Helicopter	144959.91
S18_2238	1998 Chrysler Plymouth Prowler	142530.63
S18_1749	1917 Grand Touring Sedan	140535.60
S12_2823	2002 Suzuki XREO	135767.03

Obiectiv

Care este venitul total din vânzări pentru fiecare lună și an, în ordine cronologică?

Interogare

```
SELECT
    year(o.orderDate) AS anul,
    month(o.orderDate) AS luna,
    sum(od.quantityOrdered * od.priceEach) AS VenitDinVanzari
FROM orders o
JOIN orderdetails od
    ON o.orderNumber = od.orderNumber
GROUP BY 1,2
ORDER BY 1,2;
```

Rezultat

anul	luna	VenitDinVanzari
2003	1	116692.77
2003	2	128403.64
2003	3	160517.14
2003	4	185848.59
2003	5	179435.55
2003	6	150470.77
2003	7	201940.36
2003	8	178257.11
2003	9	236697.85
2003	10	514336.21
2003	11	988025.15
2003	12	276723.25
2004	1	292385.21
2004	2	289502.84
2004	3	217691.26
2004	4	187575.77
2004	5	248325.30
2004	6	343370.74
2004	7	325563.49
2004	8	419327.09
2004	9	283799.80
2004	10	500233.86
2004	11	979291.98
2004	12	428838.17

Obiectiv

Care sunt primii 10 clienți cu cele mai mari venituri din vânzări?

Interogare

```
SELECT
    o.customerNumber,
    c.customerName,
    sum(quantityOrdered * priceEach) AS VenitTotal
FROM orderdetails od
JOIN orders o
    ON od.orderNumber = o.orderNumber
JOIN customers c
    ON o.customerNumber = c.customerNumber
GROUP BY 1,2
ORDER BY 3 DESC
LIMIT 10;
```

Rezultat

customerNumber	customerName	VenitTotal
141	Euro+ Shopping Channel	820689.54
124	Mini Gifts Distributors Ltd.	591827.34
114	Australian Collectors, Co.	180585.07
151	Muscle Machine Inc	177913.95
119	La Rochelle Gifts	158573.12
148	Dragon Souveniers, Ltd.	156251.03
323	Down Under Souveniers, Inc	154622.08
131	Land of Toys Inc.	149085.15
187	AV Stores, Co.	148410.09
450	The Sharp Gifts Warehouse	143536.27

Obiectiv

Afișează pentru fiecare an:

- **veniturile totale**
- **veniturile din anul precedent**
- **variația procentuală față de anul precedent**

Interogare

```sql
WITH tbl_VenituriAnuale AS
(
SELECT
    year(orderDate) AS an,
    SUM(quantityOrdered * priceEach) AS VenituriTotale
FROM orders o
JOIN orderdetails od
    ON o.orderNumber = od.orderNumber
GROUP BY 1
)
SELECT *, LAG(VenituriTotale, 1) OVER(ORDER BY an) AS VenituriAnPrecedent,
       ( VenituriTotale - LAG(VenituriTotale, 1) OVER(ORDER BY an) ) /
         LAG(VenituriTotale, 1) OVER(ORDER BY an) * 100 AS `Variatie%`
FROM tbl_VenituriAnuale;
```

Rezultat

an	VenituriTotale	VenituriAnPrecedent	Variatie%
2003	3317348.39	NULL	NULL
2004	4515905.51	3317348.39	36.129974
2005	1770936.71	4515905.51	-60.784460

Obiectiv

Pentru fiecare linie de produse, identifică primele 3 produse cu cele mai mari venituri din vânzări.

Interogare

```
WITH tbl_Ven_rank AS (
SELECT p.productLine, p.productCode, p.productName,
       sum(quantityOrdered * priceEach) AS VenTot,
       DENSE_RANK() OVER(PARTITION BY productLine
                         ORDER BY sum(quantityOrdered * priceEach) DESC) AS NrRank
FROM products p
JOIN orderdetails od
    ON p.productCode = od.productCode
GROUP BY 1,2,3 )
SELECT * FROM tbl_Ven_rank
WHERE NrRank <= 3;
```

Rezultat

productLine	productCode	productName	VenTot	NrRank
Classic Cars	S18_3232	1992 Ferrari 360 Spider red	276839.98	1
Classic Cars	S12_1108	2001 Ferrari Enzo	190755.86	2
Classic Cars	S10_1949	1952 Alpine Renault 1300	190017.96	3
Motorcycles	S10_4698	2003 Harley-Davidson Eagle Drag Bike	170686.00	1
Motorcycles	S12_2823	2002 Suzuki XREO	135767.03	2
Motorcycles	S10_2016	1996 Moto Guzzi 1100i	109998.82	3
Planes	S18_1662	1980s Black Hawk Helicopter	144959.91	1
Planes	S700_2834	ATA: B757-300	102786.38	2
Planes	S24_1785	1928 British Royal Navy Airplane	94885.37	3
Ships	S24_2011	18th century schooner	112427.12	1
Ships	S700_3505	The Titanic	84992.25	2
Ships	S700_3962	The Queen Mary	78919.06	3
Trains	S18_3259	Collectable Wooden Train	82617.12	1
Trains	S32_3207	1950's Chicago Surface Lines Streetcar	53791.99	2
Trains	S50_1514	1962 City of Detroit Streetcar	52123.81	3
Trucks and Buses	S12_1666	1958 Setra Bus	119085.25	1
Trucks and Buses	S24_2300	1962 Volkswagen Microbus	118774.33	2
Trucks and Buses	S18_2319	1964 Mercedes Tour Bus	117669.66	3
Vintage Cars	S18_1749	1917 Grand Touring Sedan	140535.60	1
Vintage Cars	S18_2795	1928 Mercedes-Benz SSK	132275.98	2
Vintage Cars	S18_3140	1903 Ford Model A	111528.82	3

Obiectiv: Pentru fiecare client afișează:

- valoarea totală a comenzilor
- valoarea totală a plăților
- diferența dintre comenzile totale și plățile totale

Interogare

```
WITH tbl_comenzi AS (
SELECT o.customerNumber, c.customerName,
       sum(quantityOrdered * priceEach) AS Val_Comenzi
FROM orderdetails od JOIN orders o ON od.orderNumber = o.orderNumber
    JOIN customers c ON o.customerNumber = c.customerNumber
GROUP BY 1,2 ),
tbl_plati AS (
SELECT p.customerNumber, c.customerName, sum(amount) AS Val_Plati
FROM payments p JOIN customers c
     ON p.customerNumber = c.customerNumber
GROUP BY 1,2 ),
totiClientii AS (
SELECT customerNumber, customerName FROM tbl_comenzi
UNION
SELECT customerNumber, customerName FROM tbl_plati )
SELECT t.customerNumber, t.customerName, Val_Comenzi, Val_Plati,
       coalesce(Val_Comenzi,0) - coalesce(Val_Plati,0) AS DifComenzi_Plati
FROM totiClientii t LEFT JOIN tbl_comenzi c
     ON t.customerNumber = c.customerNumber
     LEFT JOIN tbl_plati p
     ON t.customerNumber = p.customerNumber;
```

Rezultat

customerNumber	customerName	Val_Comenzi	Val_Plati	DifComenzi_Plati
141	Euro+ Shopping Channel	820689.54	715738.98	104950.56
450	The Sharp Gifts Warehouse	143536.27	59551.38	83984.89
382	Salzburg Collectables	137480.07	85060.00	52420.07
362	Gifts4AllAges.com	84340.32	33533.47	50806.85
201	UK Collectables, Ltd.	106610.72	61167.18	45443.54
448	Scandinavian Gift Ideas	120943.53	76776.44	44167.09
328	Tekni Collectables Inc.	81806.55	38281.51	43525.04
282	Souveniers And Things Co.	133907.12	91655.61	42251.51
119	La Rochelle Gifts	158573.12	116949.68	41623.44
131	Land of Toys Inc.	149085.15	107639.94	41445.21

Obiectiv: Pentru fiecare an calculează:
- Numărul total de comenzi expediate
- Numărul comenzilor expediate cu întârziere (shippedDate > requiredDate)
- Procentul comenzilor expediate cu întârziere

Interogare

```
SELECT year(orderDate) AS an,
count(*) AS TotalComenziExpediate,
sum( CASE WHEN shippedDate > requiredDate THEN 1 ELSE 0
     END) AS ComenziCuIntarziere,
sum( CASE WHEN shippedDate > requiredDate THEN 1 ELSE 0
     END) / count(*) * 100 AS Procent
FROM orders
WHERE shippedDate IS NOT NULL
GROUP BY year(orderDate);
```

Rezultat

an	TotalComenziExpediate	ComenziCuIntarziere	Procent
2003	110	1	0.9091
2004	147	0	0.0000
2005	55	0	0.0000

Obiectiv: Identifică cele mai lungi perioade consecutive de zile în care compania a înregistrat cel puțin o comandă. Afișează pentru fiecare perioadă:

- data de început
- data de sfârșit
- numărul de zile consecutive

Interogare

```sql
WITH dates AS (
SELECT DISTINCT orderDate
FROM orders ),
dates_nr AS (
SELECT orderDate, ROW_NUMBER() OVER(ORDER BY orderDate) AS row_nr
FROM dates ),
dates_dif AS (
SELECT orderDate, date_sub(orderDate, Interval row_nr Day) AS Dif
FROM dates_nr )
SELECT Dif, min(orderDate) AS min_date, max(orderDate) AS max_date,
       count(*) AS nr_zile_consec
FROM dates_dif
GROUP BY Dif
ORDER BY nr_zile_consec DESC;
```

Rezultat

Dif	min_date	max_date	nr_zile_consec
2004-05-08	2004-11-15	2004-11-25	11
2003-08-28	2003-11-04	2003-11-08	5
2003-08-30	2003-11-11	2003-11-15	5
2003-11-07	2004-02-18	2004-02-22	5
2004-05-02	2004-11-01	2004-11-05	5
2004-07-11	2005-02-28	2005-03-04	5
2003-08-12	2003-10-08	2003-10-11	4
2003-08-18	2003-10-20	2003-10-23	4
2003-09-01	2003-11-18	2003-11-21	4
2004-01-29	2004-06-14	2004-06-17	4
2004-03-25	2004-09-07	2004-09-10	4
2004-04-20	2004-10-13	2004-10-16	4
2004-05-12	2004-12-01	2004-12-04	4